

Technical Supplement: Receptiveness, Not Sycophancy: Distinguishing Engagement from Deference in Language Models

Anonymous submission

Overview

This supplement documents the measures, corpora, prompts, and experimental procedures underlying the main text, and reports supporting analyses that did not fit within the paper.

To write: *One paragraph mapping the main text's three "technical supplement" pointers to the sections here: the receptiveness scorer (S1), the rewrite prompt and its validation (S3), and the experiment instrument (S4). Also state where code and data live.*

S1. The Receptiveness Measure

The main text reports that our rubric-based score correlates with human receptiveness ratings at $r = 0.48$ in five-fold cross-validation, against $r = 0.34$ for the `politeness` package on the same texts.

To write: *The full specification of the scorer.*

- The rubric dimensions and their definitions. Five raise predicted receptiveness (hedging, emphasized agreement, acknowledgement of the other's perspective, positive reframing, invited curiosity) and five lower it (negation, dismissive limiters, explicit disagreement, interpersonal negative emotion, confrontational questioning). Give the 0–4 intensity anchors and say why intensity anchors rather than counts—this is also the answer to a length-sensitivity objection.
- The extraction prompt and the judge model used to score the dimensions.
- The calibration sample: the $n = 2,860$ human-rated texts released with [cite: Yeomans et al. replication package], and what those texts are.
- The fitted linear map, with coefficients, and the cross-validation procedure behind $r = 0.48$.
- The `politeness` baseline. Note that `receptive_train` is the package's own training data, so its $r = 0.34$ is not held out for it while ours is—the comparison runs against us.

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- The multi-rater noise ceiling and what fraction of it the score attains, so $r = 0.48$ is not read against a ceiling of 1.
- The standardization convention: all receptiveness values in the paper are in standard deviations of the human top-comment distribution.

S2. Corpora and Sycophancy Scoring

To write: *How the response corpus was built and scored.*

- The `r/AmITheAsshole` slice inherited from ELEPHANT [cite: cheng_elephant_2026]: which posts, how the top comment was selected, how many, and any filtering we applied on top.
- How model responses were generated: prompt, decoding parameters, dates, and the exact model versions behind the labels used in the figures.
- Any modification to the ELEPHANT classification prompts, and why. If the prompts are unmodified, say so explicitly—a reader comparing our rates to theirs will want to know.
- The pairwise construction of the positivity measure: each model response is scored against the corresponding human top comment, per [cite: sharma_towards_2025].
- Judge model and version for the sycophancy indicators, and any agreement check against human labels.

S3. The Receptiveness Transformation

The main text reports that the transformation raises receptiveness by 1.54 standard deviations (95% CI [1.47, 1.61]) while preserving the substantive verdict in 98.5% of cases (95% CI [97.4%, 99.2%]).

To write: *The rewrite procedure and the evidence that it preserves substance.*

- The full rewrite prompt, verbatim.
- The verdict-preservation judge: what it compares, its adjudication prompt, and how the 98.5% is computed. State that the judged pairs are the same rewrites used in Figure 2, not an earlier rewrite run.
- What the 1.5% of failures look like, and in which direction they fail.

- Length. The rewrites run longer than the originals; report the distribution of the length ratio and the length-matched replication (restricting to pairs within $\pm 10\%$ of the original’s length still yields a large increase in measured social sycophancy), so that the Figure 2 result is not attributable to verbosity.
- Naturalness or fluency checks, if reported.

S4. Experiment Details

To write: *Everything needed to replicate the study in Section 3 of the main text.*

- Preregistration: where it is, and any deviation from it.
- Recruitment, eligibility, compensation, and IRB status.
- Item selection: the pool of 100 scenarios, the requirement that rewriting raise receptiveness by at least one standard deviation, and the length and content filters. Note the consequence—selected items gain more receptiveness (2.70 SD human, 2.20 SD model) than the corpus as a whole (1.54 SD)—since a reader will otherwise notice the two numbers and wonder.
- The full instrument: item wording, response scales, and the editing applied to posts and comments (profanity, Reddit-specific language).
- Randomization: presentation order, and assignment of participants to items.
- Quality control: the attention and speed flags, how many responses each removed, and whether results are robust to keeping them.

S5. Substantive Deference and the Mitigations

To write: *The construction of $\text{Syc}(\pi)$ and of both mitigation arms.*

- How the first- and third-person presentations of each case are constructed, with an example pair, and the check that third-person prompts do not elicit first-person responses.
- The verdict extraction step and the pairwise adjudication prompt that decides which response is more favorable to the asker.
- The estimator and its inference: $\hat{T}$ is a probability-of-superiority statistic, ties split evenly, so inference reduces to a sign test on the discordant pairs. Report per-model counts.
- The receptiveness system prompt used for the prompt-level mitigation.
- The self-probing implementation used for the inference-time mitigation: the call structure, what the model sees from its own first pass, and the rewrite instruction.
- Run provenance. The prompt-level arm and the unmitigated/inference-time arms come from separate runs over the same 200 posts; say so.
- Llama 4 Scout, which the main text sets aside. Report its numbers and note that it is the only model above invariance before any intervention.

To write: *A table giving, for every model and arm, n , the discordant counts, $\hat{T}$, and mean receptiveness. This is the table the main text’s prose summarizes, and it belongs here in full.*

S6. Robustness: *The Ethicist*

To write: *The out-of-domain replication of the Figure 1 result on The Ethicist advice column. Describe the corpus, how columns and responses were paired, and whether the receptiveness/social-sycophancy association replicates. Figure S1 is the analogue of main-text Figure 1.*

[**TODO:** `plots/ethicist_corr_cont.pdf` does not exist yet—the block that writes it is in `analysis.R` but has not been run, so the figure below will not compile until it is. Re-run and copy into `manuscript/Figures/`.]

S7. Additional Figures

Social sycophancy and receptiveness, by individual measure

To write: *One paragraph: main-text Figure 1 pools the four indicators into a total score; these two figures decompose it, and the point is that no single indicator drives the association.*

Experimental outcomes in full

To write: *One paragraph: the main text reports summary statistics for quality and the two preference items; these figures give the full distributions and the splits by the participant’s own verdict.*

To write: *Decide whether the two preference-by-verdict panels (`advice_listen.pdf`, `advice_personal.pdf`) belong here as well. See the note below before including them.*

[**TODO:** In `analysis.R`, the `ggsave` block under “Pref share by verdict” writes `p_advice_verdict` to `advice_personal.pdf` but writes `p_listen`—the main-text Figure 4 panel—to `advice_listen.pdf`, rather than `p_listen_verdict`. The listen-by-verdict figure is therefore never saved. Fix before including either panel here.]

Item-level heterogeneity

To write: *One paragraph: main-text Figure 3 shows the preference gap shrinking as the original becomes more receptive; this figure shows the same heterogeneity on the quality outcome, item by item.*

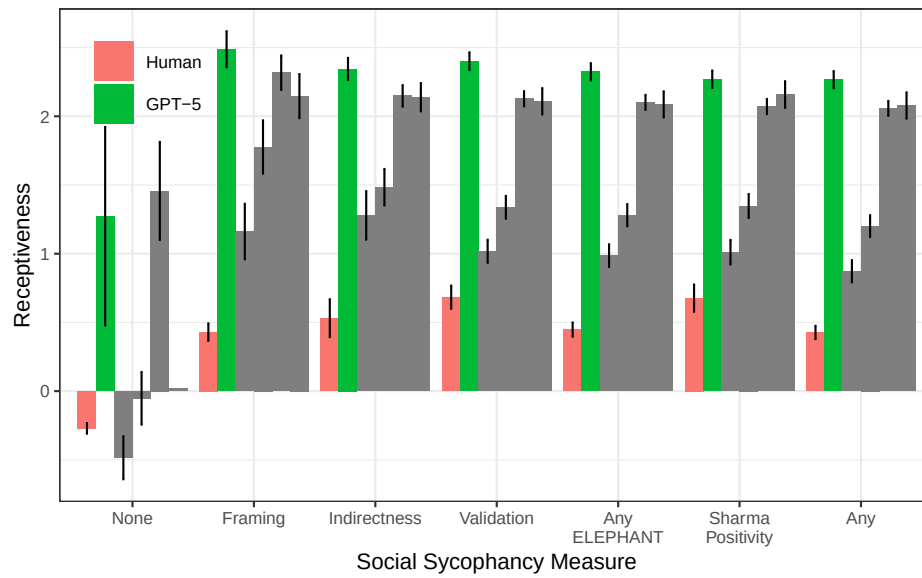


Figure S1: Receptiveness by individual social-sycophancy indicator, for human top comments and GPT-5. “None” is responses triggering no indicator; “Any ELEPHANT” pools validation, indirectness, and framing; “Sharma positivity” is the feedback-positivity indicator. Bars are ± 2 standard errors. Every indicator is associated with higher receptiveness.

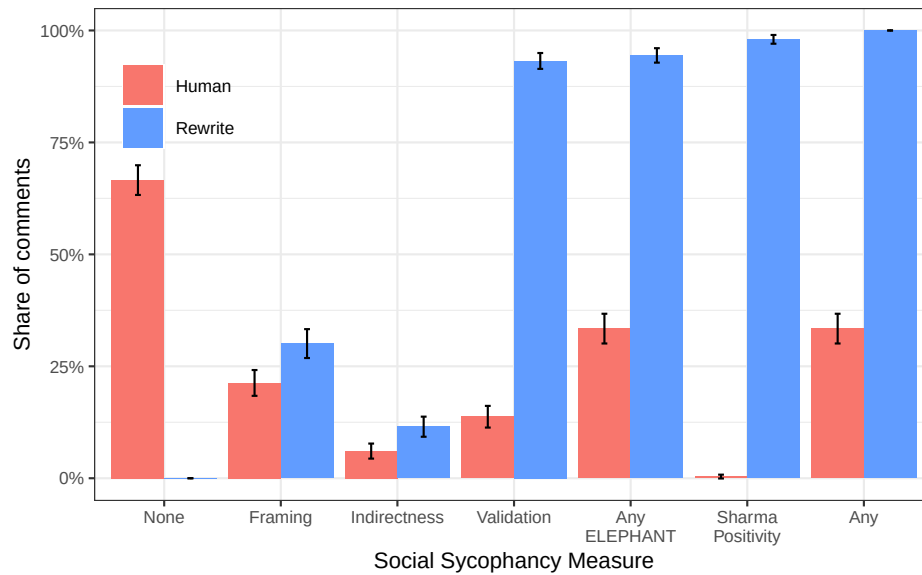


Figure S2: Share of comments triggering each social-sycophancy indicator, before and after the receptiveness-increasing rewrite. Bars are ± 2 standard errors. Per-indicator decomposition of main-text Figure 2.

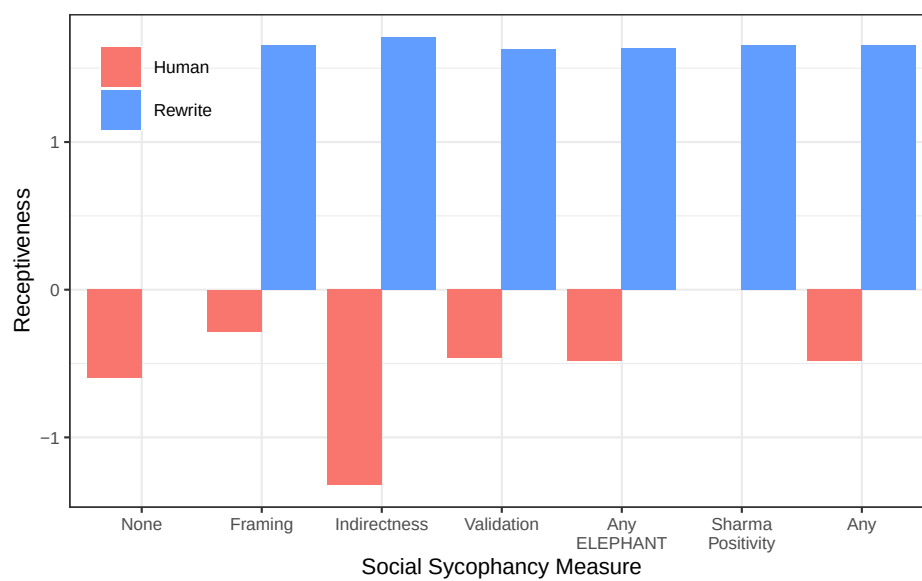


Figure S3: Receptiveness by social-sycophancy indicator, restricted to the 100 scenarios used in the experiment. The experimental subsample behaves like the full corpus.

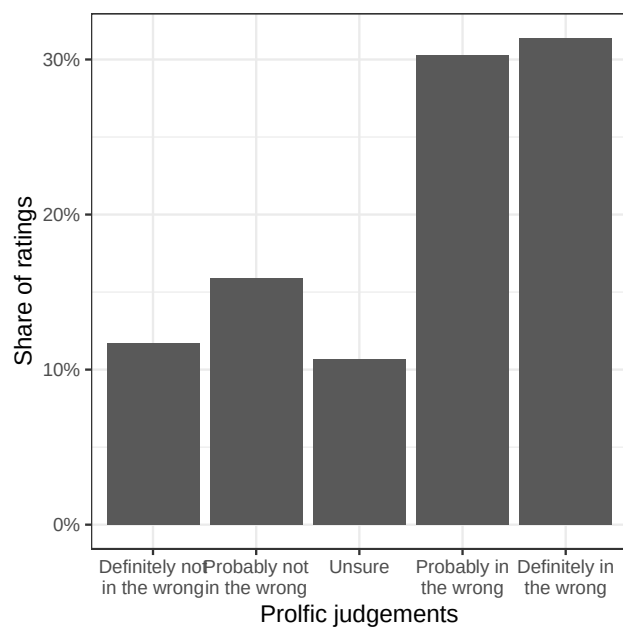


Figure S4: Distribution of participants' own verdicts on the experimental items. Although every item is drawn from the AITA-YTA slice, where both the Reddit top comment and the models judged the asker to be in the wrong, participants judged the asker not to be in the wrong on 27.6% of ratings (95% CI [24.8%, 30.5%]).

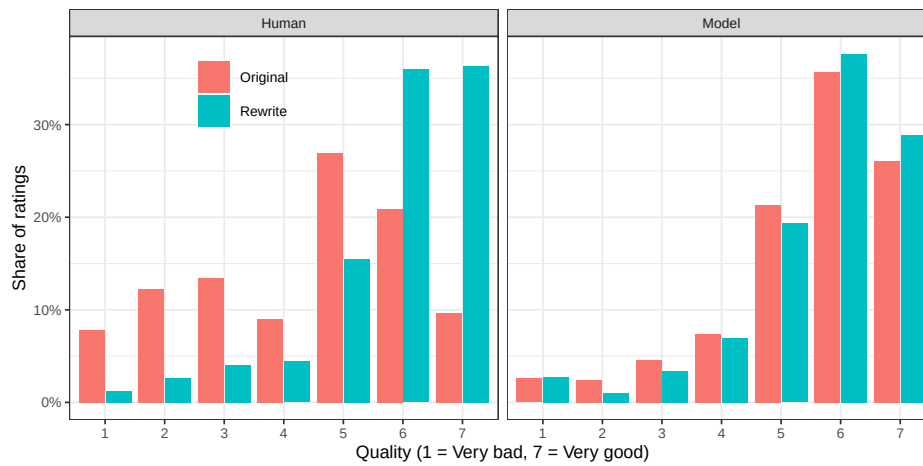


Figure S5: Full distribution of seven-point quality ratings for the original response and its receptive rewrite, faceted by whether the original was a human top comment or a model generation.

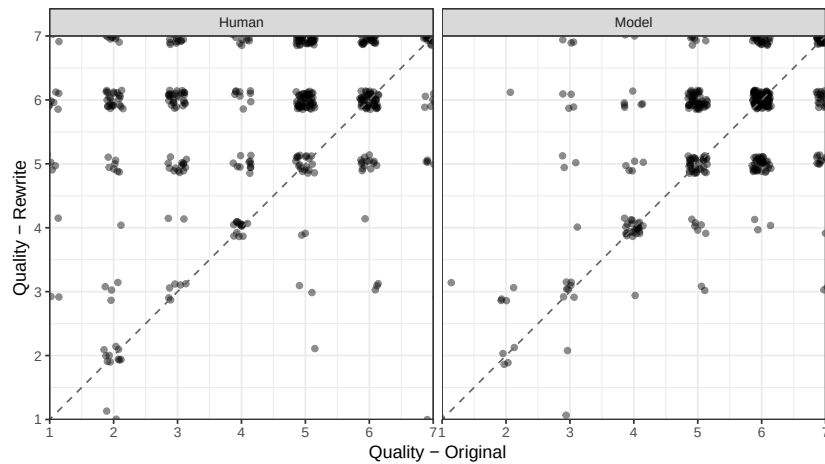


Figure S6: Paired quality ratings, rewrite against original, faceted by provenance. Points above the diagonal are ratings on which the participant preferred the rewrite.

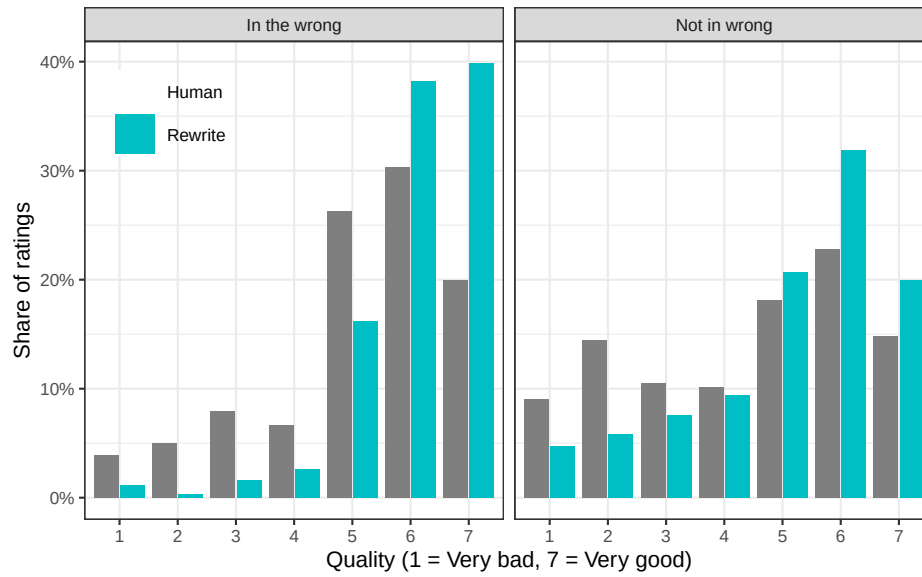


Figure S7: Quality ratings split by the participant's own verdict on the case. The rewrite is rated higher whether or not the participant judged the asker to be in the wrong, so the preference is not confined to participants sympathetic to the asker.

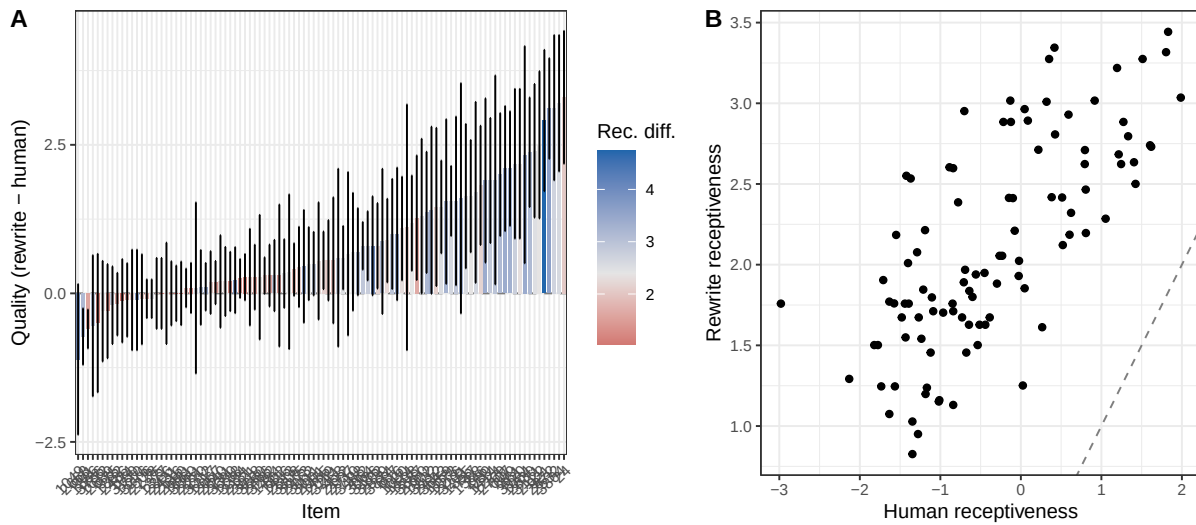


Figure S8: (A) Per-item quality difference, rewrite minus original, ordered by effect size and shaded by the item's receptiveness gain. Bars are ± 2 standard errors. (B) Receptiveness of each item's rewrite against that of its original; all items lie above the diagonal, since items were selected to gain at least one standard deviation.